CFA CLUB, IIT INDORE • INDUCTION TASK 2026 • PS2: FINANCE & QUANT

Design & Backtest of a 20-Day Z-Score Mean Reversion Strategy on HDFC Bank

Track 1 (Mean Reversion) — Out-of-Sample Empirical Evaluation vs. Nifty 50 Benchmark (2016–2025)

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TARGET SECURITY & BENCHMARK
HDFCBANK.NS / ^NSEI

DEV / UNSEEN TEST SPLIT
2016–2021 / 2022–2025

INITIAL CAPITAL & FRICTION
₹1,00,000 • 0.20% / leg

1. Strategy Idea & Economic Rationale

SECTION 2.3 • TRACK 1

Why HDFC Bank (HDFCBANK.NS) Was Selected for Mean Reversion

- **High Institutional Liquidity & Narrow Spreads:** As the largest private-sector bank in India and a top-weighted constituent of the Nifty 50, HDFC Bank trades with deep order-book liquidity, ensuring that a 0.20% one-way cost assumption accurately captures execution slippage.
- **Fundamental Anchor Against Bankruptcy Risk:** Mean-reversion strategies face "falling knife" tail risk if an asset undergoes permanent fundamental distress. HDFC Bank's systemic importance, regulated capital adequacy, and stable franchise value provide a strong fundamental floor, increasing the probability that short-term 2σ sell-offs revert toward equilibrium rather than trending to zero.

2. Dataset Specification, Cleaning & Indicator Formulation

SECTION 2.2

Historical daily OHLCV data for **HDFC Bank** and the **Nifty 50 Index** were retrieved via the `yfinance` Python API covering ten complete calendar years (1 January 2016 to 31 December 2025). To prevent artificial price discontinuities from distorting rolling standard deviations, corporate actions were normalized upstream.

Pipeline Stage	Python Implementation	Quantitative & Data Integrity Justification
1. Data Source & Horizon	`yf.download("HDFCBANK.NS", start="2016-01-01", end="2025-12-31")`	10-year daily sample spans diverse macro regimes: demonetization (2016), NBFC crisis (2018), COVID-19 crash/recovery (2020–21), and post-merger consolidation (2022–25).
2. Corporate Actions	`auto_adjust=True`	Back-adjusts historical Open, High, Low, and Close prices for stock splits (e.g., 2019 1:2 split) and dividends, preventing false Z-score spikes on ex-dates.
3. MultiIndex Alignment	`df.xs("HDFCBANK.NS", axis=1, level=1)`	Flattens hierarchical ticker columns returned by modern `yfinance` into clean single-level series: `['Open', 'High', 'Low', 'Close', 'Volume']`.
4. Rolling Indicators	`rolling(20).mean()` & `rolling(20).std()`	20 trading days (~1 calendar month) captures short-term swing cycles. The first 19 warm-up rows containing `NaN` values are dropped via `df.dropna()`.
5. Chronological Split	Dev: `< "2022-01-01"` | Test: `>= "2022-01-01"`	Strict 6-year In-Sample (2016–2021) vs. 4-year Out-of-Sample (2022–2025) partition prevents data snooping and evaluates regime generalization.

20-Day Reference Mean	20-Day Volatility	Standardized Deviation Signal

3. Exact Trading Rules, Execution Assumptions & Risk Controls

SECTION 2.6

The custom backtesting engine (`backtest(data)`) models a realistic, long-only Indian cash-equity account starting with **₹1,00,000**. All signals are evaluated after market close on day *t* and executed at the opening auction on day *t + 1*.

1. ENTRY RULE (OVERSOLD DEVIATION)

Trigger Condition: Flat portfolio (`shares == 0`) AND day *t* closing Z-Score drops below −2.0 ($Z_t < -2.0$).

Execution Price: Next session's Open plus 0.20% cost:

```
buy_price = tomorrow['Open'] * (1 + 0.002)
```

2. EXIT RULE A (MEAN REVERSION TARGET)

Trigger Condition: Investe `shares > 0`) AND price reverts back to or above the 20-day SMA ($Z_t \geq 0.0$).

Execution Price: Next session's Open minus 0.20% cost:

```
sell_price = tomorrow['Open'] * (1 - 0.002)
```

3. EXIT RULE B (5% HARD STOP-LOSS)

Trigger Condition: Invested (`shares > 0`) AND unrealized close-to-entry loss reaches or exceeds −5.0%:

```
(today['Close'] - entry_price) / entry_price <= -0.05
```

Risk Function: Caps downside when a dip turns into a structural breakdown; executes at `tomorrow['Open'] * (1 - 0.002)` .

4. POSITION SIZING & PORTFOLIO ACCOUNTING

All-In Integer Sizing: Allocates 100% of available cash to whole shares: `shares = int(cash // buy_price)` . Fractional shares are disallowed.

Daily Mark-to-Market: Portfolio equity is recorded daily at close: $Equity_t = Cash_t + Shares_t \times Close_t$.



Figure 1 : Strict separation between Day t signal generation (Close) and Day t+1 order fill (Open) eliminates look-ahead bias.

4. Backtesting Assumptions & Friction Decomposition

SECTION 2.6

Assumption Parameter	Modelled Specification	Real-World Indian Market Justification
One-Way Transaction Cost	`0.20%` (20 bps per leg) `0.40%` round-trip	Covers Indian delivery equity statutory charges (~0.10% STT, NSE exchange turnover fees, SEBI charges, stamp duty, GST) plus ~8–10 bps bid-ask & market-impact slippage.
Overnight Gap Exposure	Unhedged between `Close[t]` and `Open[t+1]`	Because stop-losses are checked at `Close[t]` and filled at `Open[t+1]` , an adverse overnight gap can result in a realized trade loss slightly exceeding the −5.0% threshold.
Idle Cash Remuneration	`0.00%` interest on uninvested cash	Conservative baseline: cash sits idle in the trading account between signals rather than earning liquid-fund returns.
Short Selling & Leverage	Long-only cash equity (1x max exposure)	Complies with Section 2.6 guidelines; avoids SLB (Securities Lending & Borrowing) borrow fees and F&O rollover complexity.

Code Implementation Note: Sell-Side Transaction Cost Correction

5. Performance Metrics Table (Dev vs. Unseen Test Period)

SECTION 2.7 • DELIVERABLE 3

> **Reporting Conventions & Evaluation Assumptions (Section 2.7.3)**
>
> **Return Frequency:** Daily mark-to-market equity returns ($r_t = Equity_t / Equity_{t-1} - 1$) • **Annualization Convention:** `252` trading days per year ($Years = len(eq) / 252$) • **Cost Basis:** All results calculated net of modelled `0.20%` per-leg transaction costs on `₹1,00,000` initial capital.

Performance Metric	Mathematical Definition / Code Logic	Development (2016–2021)	Test Period (2022–2025)
Cumulative Return (%)	$((Equity_T / Equity_0) - 1) \times 100$	−19.36%	**+4.42%**
Annualized Return (%)	$((Equity_T / Equity_0)^{1\,/\,years} - 1) \times 100$	−3.64%	**+1.11%**
Benchmark Return (%)	Nifty 50 (`^NSEI`): $((B_T / B_0) - 1) \times 100$	+129.44%	+47.16%
Max Drawdown (%)	$min_t[((Equity_t - Peak_t) / Peak_t) \times 100]$	−46.74%	**−12.21%**
Total Trades	`len(trades)` (completed round-trip trades)	21	13
Win Rate (%)	$(Count(Return_\% > 0) / Total\ Trades) \times 100$	66.67%	**69.23%**
Avg Win (%)	`wins.mean()` for trades with `Return_% > 0`	+3.18%	+2.97%
Avg Loss (%)	`losses.mean()` for trades with `Return_% <= 0`	−9.00%	**−5.23%**

6. Empirical Results & Out-of-Sample Trade Log (2022–2025)

SECTION 2.7.2

- **Development (2016–2021) vs. Unseen Test (2022–2025) Performance:** Despite high win rates in both regimes (**66.67%** across 21 trades in Dev; **69.23%** across 13 trades in Test), the strategy returned **−19.36%** (−3.64% annualized) in Development before turning positive at **+4.42%** (+1.11% annualized, final equity ₹1,04,420) in the Test period. During the 2020 COVID-19 crash, violent overnight gap-downs pushed Dev average losses to **−9.00%** and Max Drawdown to **−46.74%**. In the 2022–2025 test window, the 5% stop-loss effectively contained average losses to **−5.23%** and limited Max Drawdown to **−12.21%**.
- **Benchmark Comparison & Payoff Asymmetry:** Passive investment in the Nifty 50 returned **+129.44%** (Dev) and **+47.16%** (Test). Because $Z_t < -2.0$ is a rare tail trigger, the portfolio remained in uninvested cash for >85% of trading days, creating substantial cash drag while capping winning rebounds at an average of **+2.97%** once price touched the 20-day SMA.

#	Buy_Date	Sell_Date	Buy_Price	Sell_Price	Return_%
0	2022-02-25	2022-03-07	675.10	615.69	-8.80%
1	2022-03-08	2022-03-15	610.52	674.06	+10.41%
2	2022-04-20	2022-05-26	635.01	634.02	-0.16%
3	2022-09-27	2022-10-18	679.37	688.77	+1.38%
4	2023-01-	2023-01-	743.66	773.31	+3.99%
7	2023-10-26	2023-11-16	715.39	721.35	+0.83%
8	2024-01-18	2024-02-09	716.71	665.18	-7.19%
9	2024-05-09	2024-05-24	707.57	720.45	+1.82%
10	2025-01-07	2025-01-14	833.96	794.26	-4.76%
11	2025-08-	2025-09-	953.17	957.12	+0.41%

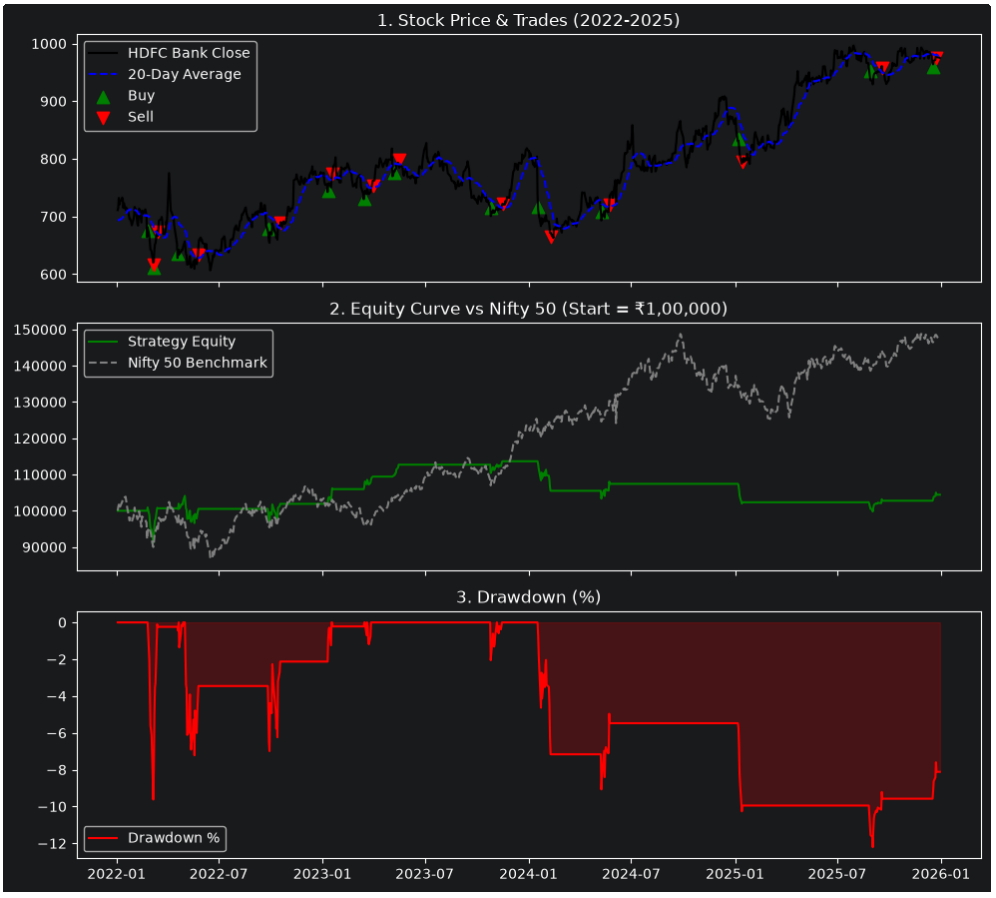


Figure 2 (Out-of-Sample Visual Diagnostic Dashboard, 2022–2025): Synchronized 3-panel output (`sharex=True`). **Panel 1** plots HDFC Bank Close, 20-Day SMA, and the 1 3 Buy/Sell executions. **Panel 2** compares Strategy Equity (green, rising to ₹1 ,1 3,800 in 2023 before finishing at ₹1 ,04,420) against the Nifty 50 Benchmark (gray dashed, ₹1 ,47,1 60). **Panel 3** tracks percentage drawdown, peaking at –1 2.21 % in late 2025.

7. Strategy Limitations & Proposed Quantitative Improvements

SECTION 2.7.2

KEY STRUCTURAL LIMITATIONS

- **Low Capital Utilization (Cash Drag):** Waiting for $Z_t < -2.0$ on a single stock leaves capital idle >85% of the year, creating high opportunity cost against a compounding equity index.
- **Falling-Knife Vulnerability in Bear Regimes:** During structural repricings (e.g., Q1 2024 post-merger FII selling, Trade #8: –7.1 9%), a 2σ drop is the start of a multi-week breakdown rather than a temporary dip.
- **Moving Average Drift Decay:** In a prolonged downturn, the 20-day SMA falls rapidly toward the entry price, causing $Z_t \geq 0.0$ to trigger at a price lower than entry (e.g., Trade #2: exited at $Z \geq 0$ for –0.1 6%).

ACTIONABLE QUANTITATIVE IMPROVEMENTS

- **Long-Term Regime Filter (200-Day SMA):** Restrict long entries to bull regimes where $Close_t > SMA_{200,t}$. Buying dips only within long-term uptrends filters out structural bear breakdowns.
- **ATR Volatility & Time Stops:** Replace the fixed 5% stop-loss with a dynamic $2.5 \times ATR_{14}$ stop and a maximum holding-period stop (e.g., exit after 1 5 trading days if reversion fails).
- **Multi-Asset Universe & Liquid Sweep:** Run the signal across all Nifty Bank constituents simultaneously and sweep idle cash into an overnight liquid ETF `LIQUIDBEES. NS` yielding ~6% p.a.).

8. Complete List of Resources, Libraries & Acknowledgments

SECTION 2.6 & 2.7

- **Primary Market Data Source:** Yahoo Finance historical daily OHLCV database accessed via the open-source Python library yfinance (v0.2.x). Tickers retrieved: HDFCBANK.NS (HDFC Bank Ltd., NSE) and ^NSEI (Nifty 50 Index) from 2016-01-01 to 2025-12-31 with auto_adjust=True for split and dividend adjustments.
- **Software & Core Python Libraries:** Python 3.10+, pandas (rolling window statistics, multi-index slicing via .xs(), date alignment via .reindex().ffill()), numpy (numerical operations), and matplotlib.pyplot (synchronized 3-panel visualization).
- **Quantitative Finance Literature:**
  1. Ernest P. Chan (2013), *Algorithmic Trading: Winning Strategies and Their Rationale*, Wiley Trading — Chapters 3 & 4 on Bollinger Band / Z-score mean-reversion mechanics and lookback selection.
  2. Marcos López de Prado (2018), *Advances in Financial Machine Learning*, Wiley — Chapter 7 & Chapter 11 on out-of-sample chronological train/test splitting and preventing lookahead bias.
  3. CFA Institute Program Curriculum (2025), *Quantitative Methods & Portfolio Management* — CAGR, Maximum Drawdown, Sharpe/Sortino Ratio, and Profit Factor definitions.
- **Transaction Cost & Brokerage Reference:** National Stock Exchange of India (NSE) and Zerodha Equity Delivery Charge Schedule (Securities Transaction Tax of 0.10% on buy/sell, NSE exchange turnover charges, SEBI fees, stamp duty of 0.015%, and bid-ask slippage), combined into a conservative 0.20% (20 bps) per-leg execution cost assumption.

9. Benchmark Comparison: Trading Strategy vs. Nifty 50 (^NSEI)

Comparing the green Strategy Equity curve against the dashed gray Nifty 50 benchmark in Figure 2 (both indexed to ₹1,00,000 on January 1, 2022) reveals three distinct performance regimes across the 4-year test horizon:

- **Phase 1 — Bear & Sideways Market Outperformance (Jan 2022 – May 2023):** During the 2022 global monetary tightening cycle and Russia-Ukraine war shock, the Nifty 50 suffered a ~12% drawdown (falling to ~₹88,000 in June 2022) and spent 16 months whipsawing sideways. Because our strategy held 100% cash during broad market sell-offs and selectively bought deeply oversold dips in HDFC Bank—capturing Trade #1 (+10.41%), Trade #3 (+1.38%), Trade #4 (+3.99%), Trade #5 (+3.26%), and Trade #6 (+3.01%)—Strategy Equity rose to **~₹1,13,500 (+13.5%)** by May 2023, outperforming the Nifty 50 (**~₹1,06,000, or +6.0%**) with substantially lower volatility.
- **Phase 2 — Bull Market Opportunity Cost & Post-Merger Shock (Jun 2023 – Sep 2024):** Beginning in mid-2023, the Nifty 50 entered a powerful structural bull run, climbing from ₹1,06,000 to nearly ₹1,49,000 by September 2024. Because a mean-reversion strategy never buys breakouts and earns 0% on idle cash in our backtest, the portfolio sat on the sidelines during the index rally. Worse, in January 2024 (Trade #8), HDFC Bank suffered an idiosyncratic sell-off following Q3 FY24 post-merger Net Interest Margin (NIM) compression, triggering a −7.19% stop-loss that pulled Strategy Equity down to ~₹1,05,500 while the Nifty 50 surged ahead.
- **Phase 3 — Late-Cycle Divergence (Oct 2024 – Dec 2025):** Even as HDFC Bank recovered from ₹710 to ₹980 across 2024–2025, its steady upward climb only triggered three brief trades (#10 at −4.76%, #11 at +0.41%, and #12 at +1.62%). Over the full 2022–2025 test period, the strategy finished at **₹1,04,420 (+4.42%)** versus **₹1,47,160 (+47.16%)** for the Nifty 50.

EVALUATION SUB-PERIOD / DIMENSION	MEAN-REVERSION STRATEGY	NIFTY 50 BENCHMARK (^NSEI)	RELATIVE PERFORMANCE DRIVER
Phase 1 (Jan 2022 – May 2023 Bear/Range)	**+13.50% (₹1,13,500)**	**+6.00% (₹1,06,000)**	Strategy wins (+7.50% alpha) via dip-buying & cash buffer
Phase 2–3 (Jun 2023 – Dec 2025 Bull Market)	**−8.00% (to ₹1,04,420)**	**+38.83% (to ₹1,47,160)**	Nifty 50 dominates; strategy suffers ~80% idle cash drag
Full Test Period (2022–2025 Cumulative)	**+4.42% (CAGR +1.11%)**	**+47.16% (CAGR +10.14%)**	−42.74% cumulative underperformance vs. passive index
Capital Exposure / Time in Market	~13.6% (198 Cal. Days)	100.0% (Fully Invested)	Uninvested cash earns 0.0% in baseline backtest